

UNIT II

Speech Recognition

Content

- Real and Complex Cepstrum
- Application of cepstral analysis to speech signal
- Feature extraction for speech
- Static and dynamic feature for speech recognition
- MFCC
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- Distance measures
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- HMM
- Robustness issues
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- Feature selection
- Gaussian Mixture model

Content

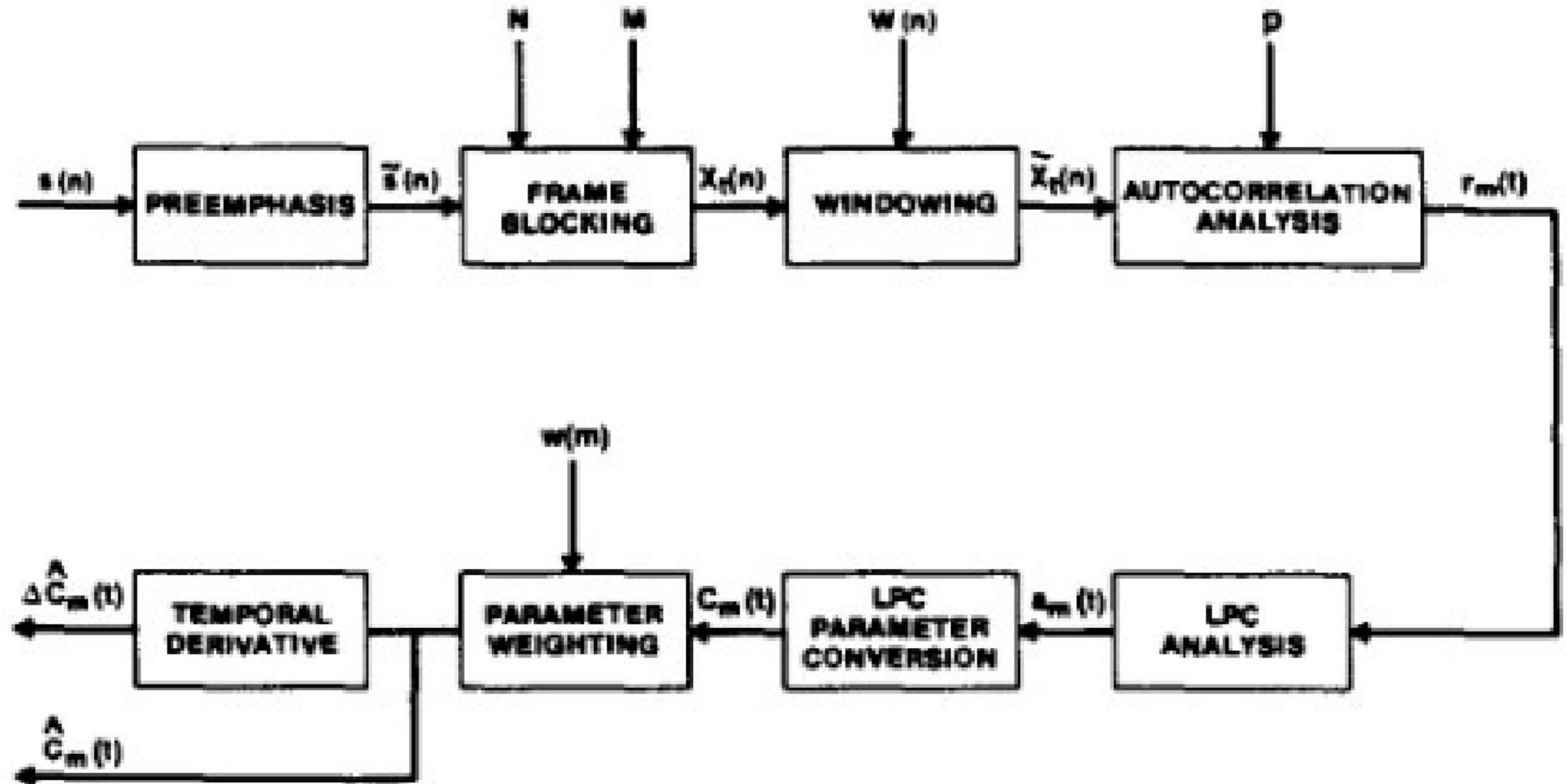
- Linear Prediction Cepstral Coefficients (LPCC)
 - Steps to produce LPCC
 - Applications of LPCC
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LPCC

- LPCC (Linear Prediction Cepstral Coefficients) is a digital feature extraction technique used in speech processing.
- It calculates the smooth spectral envelope of a vocal signal by merging Linear Predictive Coding (LPC) with Cepstral Analysis.
- MFCC models human-hearing.
- LPCC mathematically models the biological mechanism of the human vocal tract (throat, tongue, and mouth).
- It isolates the source filter (vocal cords) from the vocal tract filter (throat, tongue, mouth).

LPCC

- To extract LPCC features, the sequence of algorithmic steps:



LPCC

1. Pre-emphasis:

- The digital speech signal $s(n)$ passes through a first order FIR filter to balance the spectrum.

$$H(z) = 1 - \alpha z^{-1}, \quad 0.9 \leq \alpha \leq 1.0.$$

- It amplifying higher frequencies that naturally drop off during human speech production.
- The output of the preemphasis, $s'(n)$ of the signal $s(n)$, where $\alpha \approx 0.95$ to 0.98, is

$$s'(n) = s(n) - \alpha s(n-1)$$

LPCC

2. Frame blocking:

- Frame blocking is segmenting the speech waveform into short overlapping frames so that the signal can be treated as approximately stationary.
- The signal $s(n)$ is blocked into N number of samples with adjacent frames being separated by M samples.
- If there are L frames in speech signal and ℓ th frame of speech is denoted by $x_\ell(n)$

$$x_\ell(n) = s(M\ell + n), \quad n = 0, 1, \dots, N-1, \quad \ell = 0, 1, \dots, L-1.$$

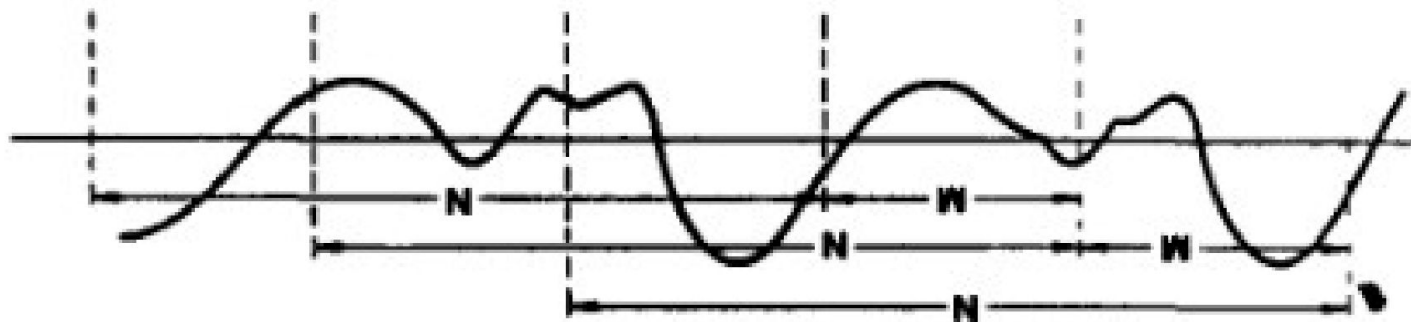


Figure 3.39 Blocking of speech into overlapping frames

LPCC

3. Windowing:

- The continuous audio is sliced into short, overlapping frames (typically 20–30 milliseconds).
- A window function (Hamming Window) is multiplied by each frame.
- It eliminates spectral leakage.
 $w(n) = 0.54 - 0.46 \cos\left(\frac{2\pi n}{N-1}\right), \quad 0 \leq n \leq N-1.$
- This process is called windowing and the resulting windowed signal is

$$\bar{x}_\ell(n) = x_\ell(n)w(n), \quad 0 \leq n \leq N-1.$$

LPCC

4. Autocorrelation analysis:

- Each windowed frame undergoes autocorrelation matching to find periodic repetitions and minimize errors between actual and predicted signal frames.

$$r_{\ell}(m) = \sum_{n=0}^{N-1-m} \tilde{x}_{\ell}(n)\tilde{x}_{\ell}(n+m), \quad m = 0, 1, \dots, p,$$

- It compute p+1 autocorrelation values for each frame.
- P is 8 to 16 generally.
- The zeroth value of the auto correlation $R_{\ell}(0)$ gives l^{th} frame energy.
- Higher lags describe short-term similarity between samples.
- These values become the input to the LPC coefficient computation.

LPCC

4. LPC analysis:

- The LPC analysis converts $p+1$ autocorrelation values into an LPC parameter set.
- Set may be LPC coefficient, reflection coefficient, log area ratio coefficients and cepstral coefficients.
- Method of converting autocorrelation coefficients to LPC parameter is called Durbin method.

$$E^{(0)} = r(0)$$

$$k_i = \left\{ r(i) - \sum_{j=1}^{i-1} \alpha_j^{(i-1)} r(|i-j|) \right\} / E^{(i-1)}, \quad 1 \leq i \leq p$$

$$\alpha_i^{(i)} = k_i$$

$$\alpha_j^{(i)} = \alpha_j^{(i-1)} - k_i \alpha_{i-j}^{(i-1)}$$

$$E^{(i)} = (1 - k_i^2) E^{(i-1)},$$

LPCC

4. LPC analysis:

- The equations are solved for $i=1\dots p$

$$a_m = \text{LPC coefficients} = \alpha_m^{(p)}, \quad 1 \leq m \leq p$$

$$k_m = \text{PARCOR coefficients}$$

$$g_m = \text{log area ratio coefficients} = \log \left(\frac{1 - k_m}{1 + k_m} \right)$$

- Using the Levinson-Durbin recursion algorithm, the system extracts linear prediction coefficients a_k .
- At each recursion, a reflection coefficient is obtained and the predictor coefficients are updated.
- The recursion also tracks the residual prediction error energy.

LPCC

5. LPC parameter conversion:

- LPC parameters set that can be derived directly from LPC coefficient are cepstral coefficients C_m via a direct recursive formula.
- The recursion used is

$$c_0 = \ln \sigma^2$$

$$c_m = a_m + \sum_{k=1}^{m-1} \left(\frac{k}{m} \right) c_k a_{m-k}, \quad 1 \leq m \leq p$$

$$c_m = \sum_{k=1}^{m-1} \left(\frac{k}{m} \right) c_k a_{m-k}, \quad m > p,$$

LPCC

5. LPC parameter conversion:

- This final conversion yields highly uncorrelated robust parameters.
- $c(m)$ derived recursively from $a_1 \dots a_p$ and gain
- The **gain term** is included in the cepstral representation is σ^2
- Q cepstral coefficients are generated from the LPC parameters, where Q may be greater than the LPC order p . ($Q \approx 3/2p$)

LPCC

6.LPC Coefficients:

- One vector per analysis frame is produced.
- Each vector can contain weighted LPC-derived cepstral coefficients.
- Temporal cepstral derivatives may be appended to capture spectral dynamics.
- The resulting feature sequence is Q cepstral coefficient and appended vector of Q cepstral time derivative .

$$V(t) = [c_1(t), c_2(t), \dots, c_Q(t), \Delta c_1(t), \Delta c_2(t), \dots, \Delta c_Q(t)]$$

Applications of LPCC

- Because LPCC accurately maps individual physical attributes of a person's voice apparatus, it excels in tasks focusing on identity and nuance over language semantics:
 1. Speaker Identification & Verification: Biometric security systems use it to recognize an individual's distinct voice print.
 2. Speech Emotion Recognition (SER): Subtle changes in throat tension due to anger, sadness, or excitement drastically affect the LPCC envelope.
 3. Speech Compression & Coding: Telecommunication networks use it to transmit high-quality speech over incredibly low bitrates.

LPCC

- Difference between MFCC and LPCC:

Metric	LPCC (Linear Prediction Cepstral Coeff.)	MFCC (Mel-Frequency Cepstral Coeff.)
Underlying Model	Mimics the biological vocal tract	Mimics the psychological human ear
Frequency Scale	Linear frequency scale	Non-linear Mel frequency scale
Noise Tolerance	Vulnerable; performance drops in noisy environments	Highly robust against background noise
Best Used For	Speaker ID, biometric security, emotion detection	Automatic Speech Recognition (ASR), translation